KAPIL BHATTARAI

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EDUCATION

M.S. in Agricultural & Biological Engineering *Jan 2025 – Present | GPA: 4.0/4.0*

University of Florida, Gainesville, FL — UF Fellowship Recipient | Thesis Advisor: Dr. Gerrit Hoogenboom

B.S. in Agriculture *2017 – 2023*

Agriculture and Forestry University (AFU), Rampur, Chitwan, Nepal

TECHNICAL SKILLS

Crop Modeling:	DSSAT v4.8, Model Calibration & Validation, Scenario & Sensitivity Analysis
Programming:	Python, R/RStudio, SQL, Git/GitHub, Shiny Apps
Data Science & ML:	Scikit-learn, TensorFlow/TFLite, Predictive Modeling, Time Series Analysis, Data Visualization
Geospatial:	QGIS, GeoPandas, Remote Sensing

EXPERIENCE

Graduate Research Assistant *Jan 2025 – Present*

Dept. of Agricultural & Biological Engineering, University of Florida — Advisor: Dr. Gerrit Hoogenboom

- Conducting thesis research on developing a process-based model for lettuce growth under hydroponics settings.
- Developing enhanced graphical visualization tool (GB2) for DSSAT v4.8.6, with modern UI and improved data processing capabilities for crop model development and large-scale data analysis.
- Provided technical guidance and DSSAT training to visiting scholars in the Crop Simulation Laboratory.

Nursery Manager *Mar 2024 – Sep 2024*

Plant Paradise — Kathmandu, Nepal

- Managed end-to-end nursery operations including plant propagation, production scheduling, inventory control, and retail sales.
- Drove digital marketing (social media, website updates) and applied data-driven sales planning to increase seasonal revenue.

Agriculture Intern *Aug 2022 – Aug 2023*

Ragolski & Son's Ltd — Netiv HaAsara, Israel

- Operated precision growing systems in high-tech greenhouses, including coco-peat substrate management, drip irrigation scheduling, and controlled-environment monitoring.
- Gained hands-on experience in hybrid vegetable variety production and commercial cut-flower operations — managing planting schedules, spraying, harvesting, packing, and quality grading.

Undergraduate Researcher & Agriculture Intern *Feb 2022 – Jul 2022*

Prime Minister Agriculture Modernization Project (PMAMP) — Chitwan, Nepal

- Precision nitrogen management in spring rice with SPAD, GreenSeeker & leaf color chart; SPAD-guided dosing cut applied N ~33% (120→80 kg/ha) with no yield loss (peer-reviewed, 2024).
- Delivered farmer training on mechanization, vermicomposting, and climate-smart agriculture practices.

SELECTED PROJECTS

DSSAT Graphical Visualization GB2 *DSSAT v4.8.6 · Data Visualization | 2025 – Present*

Enhanced graphical visualization tool for DSSAT crop model outputs, contributing to the global crop modeling research community (100+ countries).

Rogcheck — Crop Disease ID App *CNN · TensorFlow Lite · Java · Android*

Android app for offline detection of 10 crop diseases across potato, pepper bell, and tomato using a convolutional neural network.

Personal Portfolio Website *Next.js · React · CSS Modules*

Designed and built a full-stack portfolio site with dark/light theme and SEO metadata, hosted at bhattaraikapil.com.np.

CERTIFICATIONS

Certificate in Biological Systems Modeling	University of Florida	*9 credit hrs*
DSSAT 2025 International Training	University of Georgia	*May 2025*
AI Fluency: Framework & Foundations	Anthropic	*2025*
Basic Image Classification with TensorFlow	Coursera Project Network	*Apr 2022*
Remote Sensing Image Acquisition, Analysis & Applications	University of South Wales	*Mar 2022*
Data Analysis with R Programming	Coursera	*Jun 2021*
Introduction to Machine Learning	Duke University	*Apr 2021*
Working with Geospatial Data in Python	DataCamp	*Apr 2021*
Getting Started with AWS Machine Learning	AWS	*May 2021*

PRESENTATIONS & EVENTS

25th DSSAT Development Sprint — IFDC, Muscle Shoals, AL	Presenter, "Lettuce & Hydroponics Model" (In-Person)	*Aug 2026*
DSSAT 2026 Int'l Training — Univ. of Georgia	Faculty	*May 2026*
Poster Presentation — UF	Hydroponic Lettuce Growth Model	*Mar 2026*
24th DSSAT Development Sprint — Univ. of Florida	Participant	*Jan 2026*
Crop Physiology Training — Wageningen Univ. & UF	1.5 ECTS Credit	*Nov 2025*
23rd DSSAT Development Sprint — Univ. of Kentucky	Virtual Participant	*Aug 2025*
DSSAT 2025 Int'l Training — Univ. of Georgia	72 participants · 28 countries	*May 2025*
Poster Presentation — UF	DSSAT CSM Performance Evaluation	*Mar 2025*
HiPerGator Supercomputer Facility Visit — UF	UF Research Computing	*Mar 2025*
22nd DSSAT Development Sprint — UF	In-person Participant	*Jan 2025*